Gene Pulser® Electroprotocols

*** We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.**

Cell Type Mammalian, adherent

Species Used Hybrid, mouse/human, A9 fibroblast, hybrid containing human chromosomes

Molecules Electroporated DNA: linearized 8.5 kB plasmid

Before the Pulse

Cell Growth Medium DMEM + 10% Fetal Calf Serum + mycophenolic acid + xanthine

Growth Phase at Harvest 10 (7) cells / ml

Pre-pulse Incubation 10 min. on ice, (0°C)

Wash Solution HEPES Buffered Saline (HBS)

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature Room temperature

Electroporation Medium* HEPES Buffered Saline (HBS)

Cell Density Approximately 10 (7) cells / ml

Volume of Cells 0.8 ml

DNA Concentration 10 µg

DNA Resuspension Buffer Not given

Volume of DNA Not given

Cuvette Gap 0.4 cm

Voltage 0. 250, 0.450 kV

Field Strength 0.625,1.125 kV/cm

Capacitor 125, 250, 500, 960 µF

Resistor (Pulse Controller) Ω none

Time Constant Not given

After the Pulse

Outgrowth Medium Not given

Outgrowth Temperature 37 °C

Length of Incubation Not given

Selection Method or Assay Used Hygromycin B

Electroporation Efficiency Not given

Per Cent Survival Not given

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

HBS: 10mM HEPES, pH 7.2, 150 mM NaCl, 5 mM $CaCl_2$

Name of Submittor Keith A. Rosenbach

Institution Address UMDNJ New Jersey Medical School
Microbiology & Molecular Genetics
185 South Orange Ave.
Newark, NJ 07103

Survey Number 165